

Laboratorium
Multimedia dan Internet of Things
Departemen Teknik Komputer
Institut Teknologi Sepuluh Nopember

Laporan Akhir

Praktikum Jaringan Komputer

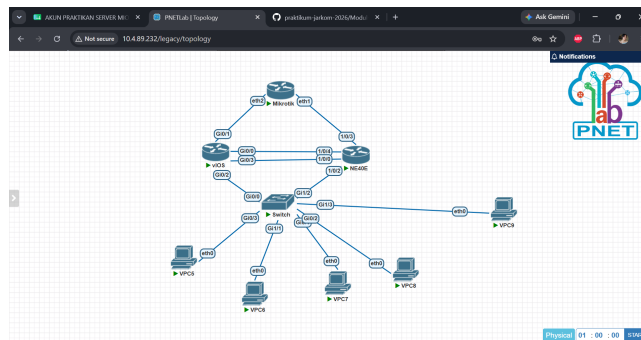
Vlan-Trunk-OSPF-MultiVendor

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05 Juni 2026

1 Langkah-Langkah Percobaan

1.1 Topologi Jaringan



Gambar 1: Topologi Sederhana

1.2 Praktikum 1 : Konfigurasi Cisco Switch

1. Masuk ke terminal Cisco Switch, membuat vlan, dan pastikan vlan sudah dibuat dan statusnya aktif

```
Switch(config-vlan)#exit
Switch(config)#
Switch(config)#vlan 30
Switch(config-vlan)#
Switch(config-vlan)#name IT
Switch(config-vlan)#
Switch(config-vlan)#exit
Switch(config)#
Switch(config)#vlan 40
Switch(config-vlan)#
Switch(config-vlan)#name MARKETING
Switch(config-vlan)#
Switch(config-vlan)#exit
Switch(config)#
Switch(config)#exit
Switch#
Switch#show
*Jun 5 02:16:22.946: %SYS-5-CONFIG_I: Configured from console by consolelan br

VLAN Name                Status    Ports
-----
1    default                active    Gi8/0, Gi8/1, Gi8/2, Gi8/3
10   FINANCE                active    Gi1/0, Gi1/1, Gi1/2, Gi1/3
20   HR                     active
30   IT                     active
40   MARKETING              active
1002 fddi-default          act/unsup
1003 tokenring-default    act/unsup
1004 fddinet-default     act/unsup
1005 trunet-default      act/unsup
Switch#
Switch#
```

Gambar 2: Reset Router

2. Konfigurasi access port untuk setiap vlan, dan konfigurasi trunk ke cisco router

```
Switch(config-if)#switchport trunk encapsulation dot1q
Switch(config-if)#
Switch(config-if)#description TRUNK-TO-CISCO-ROUTER
Switch(config-if)#
Switch(config-if)#switchport mode trunk
Switch(config-if)#
Switch(config-if)#switchport trunk allowed vlan 10,20,30,40
Switch(config-if)#
Switch(config-if)#no shutdown
Switch(config-if)#
Switch(config-if)#
Switch(config-if)#exit
Switch(config)#
Switch(config)#exit
Switch#
Switch#show
*Jun 5 02:22:07.368: %SYS-5-CONFIG_I: Configured from console by console
% Type "show ?" for a list of subcommands
Switch#
Switch#show interfaces trunk

Port      Mode      Encapsulation  Status        Native vlan
-----
Gi8/0     on        802.1q         trunking      1
Gi9/0     Vlans allowed on trunk
          10,20,30,40
Gi9/0     Vlans allowed and active in management domain
          10,20,30,40
Gi9/0     Vlans in spanning tree forwarding state and not pruned
          10,20,30,40
Switch#
Switch#
```

Gambar 3: Konfigurasi DHCP Client

1.3 Praktikum 2 : Konfigurasi Cisco Router

1. Konfigurasi interface trunk ke switch, konfigurasi vlan 10, vlan 20, vlan 30, vlan 40, konfigurasi DHCP-server untuk vlan 10, vlan 20, vlan 30, vlan 40, lalu verifikasi cisco router

2. verifikasi Cisco Router

```

*Jun  5 02:31:51.489: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to disk successfully.
Router#
Router# conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)# exit
Router#
*Jun  5 02:32:01.014: %SYS-5-CONFIG_I: Configured from console by console
Router#
Router# write memory
Building configuration...
[OK]
Router#
*Jun  5 02:32:12.963: %GRUB-5-CONFIG_WRITING: GRUB configuration is being updated on disk. Please wait...
*Jun  5 02:32:13.649: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to disk successfully.
Router#
Router# show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0      unassigned      YES unset    administratively down down
GigabitEthernet0/1      unassigned      YES unset    administratively down down
GigabitEthernet0/2      unassigned      YES unset    up
GigabitEthernet0/2.10   192.168.10.1    YES manual   up
GigabitEthernet0/2.20   192.168.20.1    YES manual   up
GigabitEthernet0/2.30   192.168.30.1    YES manual   up
GigabitEthernet0/2.40   192.168.40.1    YES manual   up
GigabitEthernet0/3      unassigned      YES unset    administratively down down
Router#
Router#
```

Gambar 4: Verifikasi Cisco Router

3. Cek DHCP pool

```

Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0      unassigned      YES unset    administratively down down
GigabitEthernet0/1      unassigned      YES unset    administratively down down
GigabitEthernet0/2      unassigned      YES unset    up
GigabitEthernet0/2.10   192.168.10.1    YES manual   up
GigabitEthernet0/2.20   192.168.20.1    YES manual   up
GigabitEthernet0/2.30   192.168.30.1    YES manual   up
GigabitEthernet0/2.40   192.168.40.1    YES manual   up
GigabitEthernet0/3      unassigned      YES unset    administratively down down
Router#
Router# show ip dhcp pool
Router#
Pool VLAN10-FINANCE :
  Utilization mark (high/low) : 100 / 0
  Subnet size (first/next) : 0 / 0
  Total addresses : 254
  Leased addresses : 1
  Pending event : none
  1 subnet is currently in the pool :
    Current index  IP address range      Leased addresses
    192.168.10.12  192.168.10.1 - 192.168.10.254  1

Pool VLAN20-HR :
  Utilization mark (high/low) : 100 / 0
  Subnet size (first/next) : 0 / 0
  Total addresses : 254
  Leased addresses : 1
  Pending event : none
  1 subnet is currently in the pool :
    Current index  IP address range      Leased addresses
    192.168.20.12  192.168.20.1 - 192.168.20.254  1
Router#
Router#
```

Gambar 5: Cek DHCP pool

```

GigabitEthernet0/3      unassigned      YES unset    administratively down down
Router#
Router# show ip dhcp pool
Router#
Pool VLAN10-FINANCE :
  Utilization mark (high/low) : 100 / 0
  Subnet size (first/next) : 0 / 0
  Total addresses : 254
  Leased addresses : 1
  Pending event : none
  1 subnet is currently in the pool :
    Current index  IP address range      Leased addresses
    192.168.10.12  192.168.10.1 - 192.168.10.254  1

Pool VLAN20-HR :
  Utilization mark (high/low) : 100 / 0
  Subnet size (first/next) : 0 / 0
  Total addresses : 254
  Leased addresses : 1
  Pending event : none
  1 subnet is currently in the pool :
    Current index  IP address range      Leased addresses
    192.168.20.12  192.168.20.1 - 192.168.20.254  1
Router#
Router# show ip dhcp binding
Router#
Bindings from all pools not associated with VRF:
IP address      Client-ID/      Lease expiration      Type
                Hardware address/
                User name
192.168.10.11    0100.5079.6668-41 Jun 06 2026 02:35 AM Automatic
192.168.20.11    0300.5079.6668-40 Jun 06 2026 02:34 AM Automatic
Router#
Router#
```

Gambar 6: IP DHCP Binding

4. Pengujian dari VPCS, show ip, PC VLAN 10 ping gateway 192.168.10.1

```

VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sourceforge.net.
For more information, please visit wiki.freecode.com.cn.
Modified version supporting unetlab by unetlab team
Press '?' to get help.

VPCS>
VPCS> ip dhcp
DHCPD IP 192.168.10.11/24 GW 192.168.10.1
VPCS> show ip
NAME      : VPCS[1]
IP/MASK    : 192.168.10.11/24
GATEWAY    : 192.168.10.1
DNS        : 8.8.8.8
DHCP SERVER : 192.168.10.1
DHCP LEASE  : 80142, 86800/43200/75600
MAC        : 08:50:79:66:68:41
IPMT      : 20000
RHOST-PORT : 127.0.0.1:30000
MTU        : 1500

VPCS> ping 192.168.10.1
84 bytes from 192.168.10.1 icmp_seq=1 ttl=255 time=2.770 ms
84 bytes from 192.168.10.1 icmp_seq=2 ttl=255 time=1.696 ms
84 bytes from 192.168.10.1 icmp_seq=3 ttl=255 time=1.437 ms
84 bytes from 192.168.10.1 icmp_seq=4 ttl=255 time=1.834 ms
84 bytes from 192.168.10.1 icmp_seq=5 ttl=255 time=2.144 ms

VPCS>
```

Gambar 7: Ping gateway 192.168.10.1 dari PC VLAN 10

5. PC VLAN 20 request ip secara dhcp, dan pastikan pc sudah dapat ip dari command, Dari PC VLAN 20 ping gateway 192.168.20.1

```

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Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.
Modified version supporting unetlab by unetlab team

Press '?' to get help.

VPCS>
VPCS> ip dhcp
DHCP IP 192.168.20.11/24 GW 192.168.20.1

VPCS> show ip
NAME      : VPCS[1]
IP/MASK   : 192.168.20.11/24
GATEWAY   : 192.168.20.1
DNS       : 8.8.8.8
DHCP SERVER : 192.168.20.1
DHCP LEASE : 86400, 86400/43200/75600
MAC       : 08:50:79:66:68:40
LPORT     : 20000
RHOST-PORT : 127.0.0.1:30000
MTU       : 1500

VPCS> ping 192.168.20.1
84 bytes from 192.168.20.1 icmp_seq=1 ttl=255 time=1.539 ms
84 bytes from 192.168.20.1 icmp_seq=2 ttl=255 time=1.770 ms
84 bytes from 192.168.20.1 icmp_seq=3 ttl=255 time=1.437 ms
84 bytes from 192.168.20.1 icmp_seq=4 ttl=255 time=1.523 ms
84 bytes from 192.168.20.1 icmp_seq=5 ttl=255 time=1.712 ms

VPCS>
```

Gambar 8: Ping gateway 192.168.20.1 dari PC VLAN 20

6. Dari PC VLAN 30 ping gateway 192.168.30.1:

```

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For more information, please visit wiki.freecode.com.cn.
Modified version supporting unetlab by unetlab team

Press '?' to get help.

VPCS>
VPCS> ip 192.168.30.10/24 192.168.30.1
Checking for duplicate address...
PCI : 192.168.30.10 255.255.255.0 gateway 192.168.30.1

VPCS> show ip
NAME      : VPCS[1]
IP/MASK   : 192.168.30.10/24
GATEWAY   : 192.168.30.1
DNS       :
MAC       : 08:50:79:66:68:3f
LPORT     : 20000
RHOST-PORT : 127.0.0.1:30000
MTU       : 1500

VPCS> ping 192.168.30.1
84 bytes from 192.168.30.1 icmp_seq=1 ttl=255 time=1.825 ms
84 bytes from 192.168.30.1 icmp_seq=2 ttl=255 time=1.485 ms
84 bytes from 192.168.30.1 icmp_seq=3 ttl=255 time=1.899 ms
84 bytes from 192.168.30.1 icmp_seq=4 ttl=255 time=1.746 ms
84 bytes from 192.168.30.1 icmp_seq=5 ttl=255 time=1.979 ms

VPCS>
```

Gambar 9: Ping gateway dari PC VLAN 30

7. Dari PC VLAN 40 ping gateway 192.168.40.1:

```

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Press '?' to get help.

VPCS>
VPCS> ip 192.168.40.10/24 192.168.40.1
Checking for duplicate address...
*PCI : 192.168.40.10 255.255.255.0 gateway 192.168.40.1

VPCS> show ip
NAME      : VPCS[1]
IP/MASK   : 192.168.40.10/24
GATEWAY   : 192.168.40.1
DNS       :
MAC       : 08:50:79:66:68:3e
LPORT     : 20000
RHOST-PORT : 127.0.0.1:30000
MTU       : 1500

VPCS> ping 192.168.40.1
84 bytes from 192.168.40.1 icmp_seq=1 ttl=255 time=1.617 ms
84 bytes from 192.168.40.1 icmp_seq=2 ttl=255 time=1.731 ms
84 bytes from 192.168.40.1 icmp_seq=3 ttl=255 time=1.644 ms
84 bytes from 192.168.40.1 icmp_seq=4 ttl=255 time=1.394 ms
84 bytes from 192.168.40.1 icmp_seq=5 ttl=255 time=1.175 ms

VPCS>
```

Gambar 10: Ping gateway dari PC VLAN 40

8. Uji ping antar-VLan dari vlan 10

```

VPCS> ping 192.168.10.1
84 bytes from 192.168.10.1: icmp_seq=1 ttl=255 time=2.778 ms
84 bytes from 192.168.10.1: icmp_seq=2 ttl=255 time=1.696 ms
84 bytes from 192.168.10.1: icmp_seq=3 ttl=255 time=1.437 ms
84 bytes from 192.168.10.1: icmp_seq=4 ttl=255 time=1.834 ms
84 bytes from 192.168.10.1: icmp_seq=5 ttl=255 time=2.144 ms

VPCS> ping 192.168.20.11
84 bytes from 192.168.20.11: icmp_seq=1 ttl=63 time=3.717 ms
84 bytes from 192.168.20.11: icmp_seq=2 ttl=63 time=3.631 ms
84 bytes from 192.168.20.11: icmp_seq=3 ttl=63 time=3.539 ms
84 bytes from 192.168.20.11: icmp_seq=4 ttl=63 time=3.868 ms
84 bytes from 192.168.20.11: icmp_seq=5 ttl=63 time=2.915 ms

VPCS> ping 192.168.30.10
84 bytes from 192.168.30.10: icmp_seq=1 ttl=63 time=1.927 ms
84 bytes from 192.168.30.10: icmp_seq=2 ttl=63 time=2.128 ms
84 bytes from 192.168.30.10: icmp_seq=3 ttl=63 time=2.899 ms
84 bytes from 192.168.30.10: icmp_seq=4 ttl=63 time=3.603 ms
84 bytes from 192.168.30.10: icmp_seq=5 ttl=63 time=3.807 ms

VPCS> ping 192.168.40.10
84 bytes from 192.168.40.10: icmp_seq=1 ttl=63 time=2.623 ms
84 bytes from 192.168.40.10: icmp_seq=2 ttl=63 time=2.809 ms
84 bytes from 192.168.40.10: icmp_seq=3 ttl=63 time=2.148 ms
84 bytes from 192.168.40.10: icmp_seq=4 ttl=63 time=3.628 ms
84 bytes from 192.168.40.10: icmp_seq=5 ttl=63 time=5.395 ms

VPCS>

```

Gambar 11: Uji ping antar VLAN

9. Masuk ke cisco router untuk melihat ip dhcp binding

```

Leased addresses : 1
Pending event : none
1 subnet is currently in the pool :
Current index IP address range Leased addresses
192.168.10.12 192.168.10.254 1

Pool VLAN20-WF :
Utilization mark (high/low) : 100 / 0
Subnet size (first/next) : 0 / 0
Total addresses : 254
Leased addresses : 1
Pending event : none
1 subnet is currently in the pool :
Current index IP address range Leased addresses
192.168.20.12 192.168.20.1 1

Router#
Router# show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address Client-ID/ Hardware address/ Lease expiration Type
192.168.10.11 0100.5079.6668-41 Jun 06 2026 02:35 AM Automatic
192.168.20.11 0100.5079.6668-40 Jun 06 2026 02:34 AM Automatic
Router#
Router# show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address Client-ID/ Hardware address/ Lease expiration Type
192.168.10.11 0100.5079.6668-41 Jun 06 2026 02:35 AM Automatic
192.168.20.11 0100.5079.6668-40 Jun 06 2026 02:34 AM Automatic
Router#
Router#

```

Gambar 12: Show ip DHCP binding dari cisco router

1.4 Praktikum 3 : Konfigurasi Huawei Router dan IP Address Antar Router

1. Konfigurasi Lan Huawei, IP VPC Lan Huawei

```

0:VPCS
Welcome to Virtual PC Simulator, version 1.0 (0.8c)
Dedicated to Daling.
Build time: Dec 31 2016 01:22:17
Copyright (c) 2007-2015, Paul Meng (mirmah@gmail.com)
All rights reserved.

VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.
Modified version supporting uerlab by uerlab team

Press '?' to get help.

VPCS>
VPCS> ip 192.168.50.10/24 192.168.50.1
Checking for duplicate address...
PC1 - 192.168.50.10 255.255.255.0 gateway 192.168.50.1

VPCS> show ip

NAME : VPCS[1]
IP/MASK : 192.168.50.10/24
GATEWAY : 192.168.50.1
DNS :
MAC : 00:50:79:66:68:42
LPORT : 28880
RHOST-PORT : 127.0.0.1:30000
MTU : 1500

VPCS>

```

Gambar 13: Verifikasi Show IP

```

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For more information, please visit wiki.freecode.com.cn.
Modified version supporting unetlab by unetlab team

Press '?' to get help.

VPCS>
VPCS> ip 192.168.50.10/24 192.168.50.1
Checking for duplicate address...
PCI : 192.168.50.10 255.255.255.0 gateway 192.168.50.1

VPCS> show ip

NAME      : VPCS[1]
IP/MASK   : 192.168.50.10/24
GATEWAY   : 192.168.50.1
DNS       :
MAC       : 00:50:79:66:68:42
LPORT     : 20000
RHOST-PORT : 127.0.0.1:30000
MTU       : 1500

VPCS> ping 192.168.50.1

84 bytes from 192.168.50.1: icmp_seq=1 ttl=255 time=0.238 ms
84 bytes from 192.168.50.1: icmp_seq=2 ttl=255 time=0.597 ms
84 bytes from 192.168.50.1: icmp_seq=3 ttl=255 time=1.747 ms
84 bytes from 192.168.50.1: icmp_seq=4 ttl=255 time=1.670 ms
84 bytes from 192.168.50.1: icmp_seq=5 ttl=255 time=1.640 ms

VPCS>
```

Gambar 14: Ping Gateway Huawei menggunakan 192.168.50.1

2. Konfigurasi IP Link cisco <-> Huawei link 1, Huawei Link 2, mikrotik

Konfigurasi IP link huawei <-> Mikrotik

Lalu verifikasi interface

```

sisko
Router(config-if)#ip address 10.10.30.1 255.255.255.252
Router(config-if)#
Router(config-if)#no shutdown
Router(config-if)#
Router(config-if)#exit
Router(config)#
Router#
*Jun  5 02:56:22.471: %LINK-3-UPDOWN: Interface GigabitEthernet0/1, changed state to up
*Jun  5 02:56:22.806: %SYS-5-CONFIG-I: Configured from console by console
*Jun  5 02:56:23.471: %LINEPROTO-5-UPDOWN: Line protocol on interface GigabitEthernet0/1, changed state to up
Router#
Router#write memory
Building configuration...
[OK]
Router#
Router#
*Jun  5 02:56:29.155: %GRUB-5-CONFIG_WRITING: GRUB configuration is being updated on disk. Please wait...
*Jun  5 02:56:29.843: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to disk successfully.
Router#
Router#show ip interface brief
Interface                                IP-Address      OK? Method Status          Protocol
GigabitEthernet0/0                      10.10.11.1      YES manual up              up
GigabitEthernet0/1                      10.10.30.1      YES manual up              up
GigabitEthernet0/2                      unassigned      YES unset up              up
GigabitEthernet0/2.10                   192.168.10.1    YES manual up              up
GigabitEthernet0/2.20                   192.168.20.1    YES manual up              up
GigabitEthernet0/2.30                   192.168.30.1    YES manual up              up
GigabitEthernet0/2.40                   192.168.40.1    YES manual up              up
GigabitEthernet0/3                      10.10.10.1      YES manual up              up
Router#
Router#
```

Gambar 15: Verifikasi Interface

```

huawei
[*HUAWEI-Ethernet1/0/3]
[*HUAWEI-Ethernet1/0/3]undo shutdown
[*HUAWEI-Ethernet1/0/3]
[*HUAWEI-Ethernet1/0/3]quit
[*HUAWEI]
[*HUAWEI]commit
[*HUAWEI]
[*HUAWEI]display ip interface brief
*down: administratively down
!down: FIB overload down
^down: standby
(C): loopback
(S): spoofing
(d): Dampeaning Suppressed
(E): E-Trunk down
The number of interface that is UP in Physical is 23
The number of interface that is DOWN in Physical is 0
The number of interface that is UP in Protocol is 16
The number of interface that is DOWN in Protocol is 7

Interface                                IP Address/Mask  Physical  Protocol VPN
Ethernet1/0/0                            10.10.10.2/30    up        up      --
Ethernet1/0/0.4094                       128.1.138.137/16 up        down    --
Ethernet1/0/1                            128.1.138.137/16 up        up      L3vpn
Ethernet1/0/2                            192.168.50.1/24  up        up      --
Ethernet1/0/2.4094                       128.1.138.137/16 up        up      L3vpn
Ethernet1/0/3                            10.10.20.1/30    up        up      --
Ethernet1/0/3.4094                       128.1.138.137/16 up        up      L3vpn
Ethernet1/0/4                            10.10.11.2/30    up        up      --
Ethernet1/0/4.4094                       128.1.138.137/16 up        up      L3vpn
Ethernet1/0/5                            unassigned       up        down    --
---- More ----
```

Gambar 16: IP sudah sesuai dan statusnya UP

```

mikrotik
MikroTik RouterOS 6.49.17 (c) 1999-2024 http://www.mikrotik.com/

Do you want to see the software license? [Y/n]: n
[?] Gives the list of available commands
command [?] Gives help on the command and list of arguments

[Tab] Completes the command/word. If the input is ambiguous,
a second [Tab] gives possible options

/ Move up to base level
.. Move up one level
/command Use command at the base level
Jun/05/2026 02:13:59 system,error,critical router was rebooted without proper shutdown
Change your password
new password>

[admin@MikroTik] > /ip address add address=10.10.20.2/30 interface=ether1 comment=TO-HUAWEI
[admin@MikroTik] > /ip address add address=10.10.30.2/30 interface=ether2 comment=TO-CISCO
[admin@MikroTik] > /ip address print
Flags: X - disabled, I - invalid, D - dynamic
# ADDRESS NETWORK INTERFACE
0 ;; TO-HUAWEI 10.10.20.0 ether1
1 ;; TO-CISCO 10.10.30.0 ether2
[admin@MikroTik] >

```

Gambar 17: Mengecek IP Address

3. Verifikasi ping antar router

```

sisco
[OK]
Router#
Router#
*Jun  5 02:56:29.155: %GRUB-5-CONFIG_WRITING: GRUB configuration is being updated on disk. Please wait...
*Jun  5 02:56:29.843: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to disk successfully.
Router#
Router#show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 10.10.11.1 YES manual up
GigabitEthernet0/1 10.10.30.1 YES manual up
GigabitEthernet0/2 unassigned YES unset up
GigabitEthernet0/2.10 192.168.10.1 YES manual up
GigabitEthernet0/2.20 192.168.20.1 YES manual up
GigabitEthernet0/2.30 192.168.30.1 YES manual up
GigabitEthernet0/2.40 192.168.40.1 YES manual up
GigabitEthernet0/3 10.10.10.1 YES manual up
Router#
Router#ping 10.10.10.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 1/1/2 ms
Router#ping 10.10.11.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.11.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 1/1/3 ms
Router#ping 10.10.30.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.30.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 1/1/1 ms
Router#

```

Gambar 18: Verifikasi dari Cisco Router

```

huawei
Ethernet1/0/8.4094 128.1.138.137/16 up up l3vpn
Ethernet1/0/9 unassigned up down --
[~HUAWEI]ping 10.10.10.1
PING 10.10.10.1: 56 data bytes, press CTRL_C to break
Reply from 10.10.10.1: bytes=56 Sequence=1 ttl=255 time=2 ms
Reply from 10.10.10.1: bytes=56 Sequence=2 ttl=255 time=1 ms
Reply from 10.10.10.1: bytes=56 Sequence=3 ttl=255 time=1 ms
Reply from 10.10.10.1: bytes=56 Sequence=4 ttl=255 time=1 ms
Reply from 10.10.10.1: bytes=56 Sequence=5 ttl=255 time=2 ms

--- 10.10.10.1 ping statistics ---
5 packet(s) transmitted
5 packet(s) received
0.00% packet loss
round-trip min/avg/max = 1/1/2 ms

[~HUAWEI]ping 10.10.11.1
PING 10.10.11.1: 56 data bytes, press CTRL_C to break
Reply from 10.10.11.1: bytes=56 Sequence=1 ttl=255 time=2 ms
Reply from 10.10.11.1: bytes=56 Sequence=2 ttl=255 time=1 ms
Reply from 10.10.11.1: bytes=56 Sequence=3 ttl=255 time=1 ms
Reply from 10.10.11.1: bytes=56 Sequence=4 ttl=255 time=1 ms
Reply from 10.10.11.1: bytes=56 Sequence=5 ttl=255 time=1 ms

--- 10.10.11.1 ping statistics ---
5 packet(s) transmitted
5 packet(s) received
0.00% packet loss
round-trip min/avg/max = 1/1/2 ms

[~HUAWEI]
[~HUAWEI]

```

Gambar 19: Verifikasi ping dari Router Huawei

```
huawei
[~HUAWEI]ping 10.10.11.1
PING 10.10.11.1: 56 data bytes, press CTRL_C to break
Reply from 10.10.11.1: bytes=56 Sequence=1 ttl=255 time=2 ms
Reply from 10.10.11.1: bytes=56 Sequence=2 ttl=255 time=1 ms
Reply from 10.10.11.1: bytes=56 Sequence=3 ttl=255 time=1 ms
Reply from 10.10.11.1: bytes=56 Sequence=4 ttl=255 time=1 ms
Reply from 10.10.11.1: bytes=56 Sequence=5 ttl=255 time=1 ms

--- 10.10.11.1 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 1/1/2 ms

[~HUAWEI]ping 10.10.20.2
PING 10.10.20.2: 56 data bytes, press CTRL_C to break
Reply from 10.10.20.2: bytes=56 Sequence=1 ttl=64 time=1 ms
Reply from 10.10.20.2: bytes=56 Sequence=2 ttl=64 time=1 ms
Reply from 10.10.20.2: bytes=56 Sequence=3 ttl=64 time=1 ms
Reply from 10.10.20.2: bytes=56 Sequence=4 ttl=64 time=1 ms
Reply from 10.10.20.2: bytes=56 Sequence=5 ttl=64 time=1 ms

--- 10.10.20.2 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 1/1/1 ms

[~HUAWEI]
[~HUAWEI]
```

Gambar 20: Verifikasi ping dari Router Huawei

```
mikrotik
Flags: X - disabled, I - invalid, D - dynamic
# ADDRESS NETWORK INTERFACE
0 ;; TO-HUAWEI
10.10.20.2/30 10.10.20.0 ether1
1 ;; TO-CISCO
10.10.30.2/30 10.10.30.0 ether2
[admin@MikroTik] > ping 10.10.20.1
SEQ HOST SIZE TTL TIME STATUS
0 10.10.20.1 56 255 1ms
1 10.10.20.1 56 255 0ms
2 10.10.20.1 56 255 0ms
3 10.10.20.1 56 255 0ms
4 10.10.20.1 56 255 0ms
5 10.10.20.1 56 255 0ms
6 10.10.20.1 56 255 0ms
7 10.10.20.1 56 255 0ms
8 10.10.20.1 56 255 0ms
9 10.10.20.1 56 255 0ms
10 10.10.20.1 56 255 0ms
sent=11 received=11 packet-loss=0% min-rtt=0ms avg-rtt=0ms max-rtt=1ms

[admin@MikroTik] > ping 10.10.30.1
SEQ HOST SIZE TTL TIME STATUS
0 10.10.30.1 56 255 0ms
1 10.10.30.1 56 255 1ms
2 10.10.30.1 56 255 1ms
3 10.10.30.1 56 255 0ms
4 10.10.30.1 56 255 1ms
5 10.10.30.1 56 255 2ms
6 10.10.30.1 56 255 0ms
sent=7 received=7 packet-loss=0% min-rtt=0ms avg-rtt=0ms max-rtt=2ms

[admin@MikroTik] >
```

Gambar 21: Verifikasi Ping dari Router Mikrotik

1.5 Praktikum 4 : Konfigurasi OSPF Multivendor dan OSPF Cost

4. Konfigurasi OSPF pada Cisco Router

```
Router#show ip protocols
*** IP Routing is NSF aware ***

Routing Protocol is "ospf 1"
  Sending updates every 30 seconds
  Invalid after 0 seconds, hold down 0, flushed after 0
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Maximum path: 16
  Routing for Networks:
    Routing Information Sources:
      Gateway         Distance      Last Update
    Distance: (default is 1)
  Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 1.1.1.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 16
  Routing for Networks:
    10.10.0.0/24 area 0
    10.10.11.0/24 area 0
    10.10.20.0/24 area 0
    192.168.10.0/24 area 0
    192.168.20.0/24 area 0
    192.168.30.0/24 area 0
    192.168.40.0/24 area 0
  Routing Information Sources:
    Gateway         Distance      Last Update
    Distance: (default is 110)

Router#
```

Gambar 22: Konfigrasi OSPF pada Cisco Router

5. Konfigurasi OSPF pada Huawei Router


```
0.80% packet loss
round-trip min/avg/max = 1/1/5 ms

[~HUAWEI]
[~HUAWEI]display ospf peer
(N) Indicates MADM neighbor

OSPF Process 1 with Router ID 2.2.2.2
  Neighbors
Area 0.0.0.0 interface 10.10.10.2 (Eth1/0/0)'s neighbors
Router ID: 1.1.1.1      Address: 10.10.10.1
State: Full           Mode: Master is Slave      Priority: 1
DR: 10.10.10.1        BDR: 10.10.10.2          MTU: 1500
Dead timer due in 37 sec
Retrans timer interval: 5
Neighbor is up for 00h00m38s
Neighbor Up Time : 2026-06-05 03:20:50
Authentication Sequence: [ 0 ]

Area 0.0.0.0 interface 10.10.11.2 (Eth1/0/4)'s neighbors
Router ID: 1.1.1.1      Address: 10.10.11.1
State: Full           Mode: Master is Slave      Priority: 1
DR: 10.10.11.1        BDR: 10.10.11.2          MTU: 1500
Dead timer due in 36 sec
Retrans timer interval: 5
Neighbor is up for 00h00m31s
Neighbor Up Time : 2026-06-05 03:20:50
Authentication Sequence: [ 0 ]

[~HUAWEI]
```

Gambar 23: Konfigurasi OSPF pada Huawei Router

```
public_Routing Table : OSPF
  Destinations : 9          Routes : 14
OSPF routing table status : <Active>
  Destinations : 5          Routes : 10
Destination/Mask  Proto  Pre  Cost  Flags NextHop  Interface
10.10.10.0/30    OSPF   10   110    D  10.10.11.1    Ethernet1/0/4
10.10.10.0/30    OSPF   10   110    D  10.10.10.1    Ethernet1/0/0
192.168.10.0/24  OSPF   10   11    D  10.10.11.1    Ethernet1/0/4
192.168.10.0/24  OSPF   10   11    D  10.10.10.1    Ethernet1/0/0
192.168.20.0/24  OSPF   10   11    D  10.10.11.1    Ethernet1/0/4
192.168.20.0/24  OSPF   10   11    D  10.10.10.1    Ethernet1/0/0
192.168.30.0/24  OSPF   10   11    D  10.10.11.1    Ethernet1/0/4
192.168.30.0/24  OSPF   10   11    D  10.10.10.1    Ethernet1/0/0
192.168.40.0/24  OSPF   10   11    D  10.10.11.1    Ethernet1/0/4
192.168.40.0/24  OSPF   10   11    D  10.10.10.1    Ethernet1/0/0

OSPF routing table status : <Inactive>
  Destinations : 4          Routes : 4
Destination/Mask  Proto  Pre  Cost  Flags NextHop  Interface
10.10.10.0/30    OSPF   10   10     D  10.10.10.2    Ethernet1/0/0
10.10.11.0/30    OSPF   10   10     D  10.10.11.2    Ethernet1/0/4
10.10.20.0/30    OSPF   10   100    D  10.10.20.1    Ethernet1/0/3
192.168.50.0/24  OSPF   10   1     D  192.168.50.1  Ethernet1/0/2

[~HUAWEI]
[~HUAWEI]
[~HUAWEI]
```

Gambar 24: IP Routing table

```
10.10.10.0/30    OSPF   10   10     D  10.10.10.2    Ethernet1/0/0
10.10.11.0/30    OSPF   10   10     D  10.10.11.2    Ethernet1/0/4
10.10.20.0/30    OSPF   10   100    D  10.10.20.1    Ethernet1/0/3
192.168.50.0/24  OSPF   10   1     D  192.168.50.1  Ethernet1/0/2

[~HUAWEI]
[~HUAWEI]
[~HUAWEI]display ospf routing

OSPF Process 1 with Router ID 2.2.2.2
  Routing Tables

Routing for Network
Destination      Cost    Type    NextHop  AdvRouter  Area
10.10.10.0/30   10      Direct  10.10.10.2  2.2.2.2    0.0.0.0
10.10.11.0/30   10      Direct  10.10.11.2  2.2.2.2    0.0.0.0
10.10.20.0/30   100     Direct  10.10.20.1  2.2.2.2    0.0.0.0
10.10.30.0/30   110     Stub    10.10.11.1  1.1.1.1    0.0.0.0
10.10.30.0/30   110     Stub    10.10.10.1  1.1.1.1    0.0.0.0
192.168.10.0/24  11      Stub    10.10.11.1  1.1.1.1    0.0.0.0
192.168.10.0/24  11      Stub    10.10.10.1  1.1.1.1    0.0.0.0
192.168.20.0/24  11      Stub    10.10.11.1  1.1.1.1    0.0.0.0
192.168.20.0/24  11      Stub    10.10.10.1  1.1.1.1    0.0.0.0
192.168.30.0/24  11      Stub    10.10.11.1  1.1.1.1    0.0.0.0
192.168.30.0/24  11      Stub    10.10.10.1  1.1.1.1    0.0.0.0
192.168.40.0/24  11      Stub    10.10.11.1  1.1.1.1    0.0.0.0
192.168.40.0/24  11      Stub    10.10.10.1  1.1.1.1    0.0.0.0
192.168.50.0/24  1       Direct  192.168.50.1  2.2.2.2    0.0.0.0

Total Nets: 9
Intra Area: 9 Inter Area: 0 ASE: 0 NSSA: 0

---- More ----
```

Gambar 25: Display OSPF Routing

6. Konfigurasi OSPF pada Mikrotik

```
0 10.10.10.1      56 255 0ms
10.10.10.20.1    56 255 0ms
sent=11 received=11 packet-loss=0% min-rtt=0ms avg-rtt=0ms max-rtt=0ms

[Admin@Mikrotik] > ping 10.10.10.1
SEQ HOST      SIZE TTL TIME  STATUS
0 10.10.10.1  56 255 0ms  Succeeded
1 10.10.10.1  56 255 0ms  Succeeded
2 10.10.10.1  56 255 0ms  Succeeded
3 10.10.10.1  56 255 0ms  Succeeded
4 10.10.10.1  56 255 0ms  Succeeded
5 10.10.10.1  56 255 0ms  Succeeded
6 10.10.10.1  56 255 0ms  Succeeded
sent=7 received=7 packet-loss=0% min-rtt=0ms avg-rtt=0ms max-rtt=0ms

[Admin@Mikrotik] > /routing ospf instance set default router-id=3.3.3.3
[Admin@Mikrotik] > /routing ospf
[Admin@Mikrotik] /routing ospf> network add network=10.10.20.0/30 area=backbone
[Admin@Mikrotik] /routing ospf> network add network=10.10.30.0/30 area=backbone
[Admin@Mikrotik] /routing ospf> /
[Admin@Mikrotik] > /routing ospf interface
[Admin@Mikrotik] /routing ospf interface> add interface=ether1 cost=100
[Admin@Mikrotik] /routing ospf interface> add interface=ether2 cost=100
[Admin@Mikrotik] /routing ospf interface> /
[Admin@Mikrotik] > /routing ospf neighbor print
0 instance=default router-id=1.1.1.1 neighbor=10.10.10.1 interface=ether2 priority=1 dr-address=10.10.10.1
backbone=10.10.30.2 state=Full state-change=5 li-retransmit=0 li-request=0 dr-kummarjoo=0
adjacency=12s
1 instance=default router-id=2.2.2.2 address=10.10.20.1 interface=ether1 priority=1 dr-address=10.10.20.1
backbone=10.10.20.2 state=Full state-change=5 li-retransmit=0 li-request=0 dr-kummarjoo=0
adjacency=12s
[Admin@Mikrotik] >
```

Gambar 26: Konfigurasi OSPF pada Mikrotik

```
sonet-7 received packet-10000% nla-rtt-0ms nsp-rtt-0ms nmr-rtt-2ms

[admin@Mikrotik] /> /routing ospf instance set default router-id 3.3.3.3
[admin@Mikrotik] /> /routing ospf
[admin@Mikrotik] /> /routing ospf network add network 10.10.20.0/30 area backbone
[admin@Mikrotik] /> /routing ospf network add network 10.10.30.0/30 area backbone
[admin@Mikrotik] /> /routing ospf /
[admin@Mikrotik] /> /routing ospf interface
[admin@Mikrotik] /> /routing ospf interface add interface-ether1 cost 100
[admin@Mikrotik] /> /routing ospf interface add interface-ether2 cost 100
[admin@Mikrotik] /> /routing ospf interface /
[admin@Mikrotik] /> /routing ospf neighbor print
0 instance-default router-id 1.1.1.1 address 10.10.30.1 interface-ether2 priority 1 dr-address 10.10.30.1
  backup-dr-address 10.10.30.2 state="Full" state-changes=5 ls-retransmits=0 ls-requests=0 db-summary=0
  adjacency=125
1 instance-default router-id 2.2.2.2 address 10.10.20.1 interface-ether1 priority 1 dr-address 10.10.20.1
  backup-dr-address 10.10.20.2 state="Full" state-changes=5 ls-retransmits=0 ls-requests=0 db-summary=0
  adjacency=125
[admin@Mikrotik] /> /ip route print where ospf
Flags: X - disabled, A - active, D - dynamic, C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
  0 - blackhole, U - unreachable, P - prohibit
# DST-ADDRESS      PREF-SRC      GATEWAY      DISTANCE
0 Ado 10.10.10.0/30      10.10.20.1      110
1 Ado 10.10.11.0/30      10.10.20.1      110
2 Ado 192.168.10.0/24     10.10.30.1      110
3 Ado 192.168.20.0/24     10.10.30.1      110
4 Ado 192.168.30.0/24     10.10.30.1      110
5 Ado 192.168.40.0/24     10.10.30.1      110
6 Ado 192.168.50.0/24     10.10.30.1      110
[admin@Mikrotik] />
```

Gambar 27: Konfigurasi OSPF pada Mikrotik

7. Verifikasi OSPF Neighbor

```
Distance: (default is 4)

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 1.1.1.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    10.10.0.0 0.0.0.3 area 0
    10.10.11.0 0.0.0.3 area 0
    10.10.30.0 0.0.0.3 area 0
    192.168.10.0 0.0.0.255 area 0
    192.168.20.0 0.0.0.255 area 0
    192.168.30.0 0.0.0.255 area 0
    192.168.40.0 0.0.0.255 area 0
  Routing Information Sources:
    Gateway         Distance      Last Update
  Distance: (default is 110)

Router#
*Jun  5 03:20:38.246: NSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on GigabitEthernet0/3 from LOADING to FULL, Loading Done
*Jun  5 03:20:45.049: NSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on GigabitEthernet0/0 from LOADING to FULL, Loading Done
Router#
*Jun  5 03:27:38.478: NSPF-5-ADJCHG: Process 1, Nbr 3.3.3.3 on GigabitEthernet0/1 from LOADING to FULL, Loading Done
*Jun  5 03:28:01.463: NSPF-5-ADJCHG: Process 1, Nbr 3.3.3.3 on GigabitEthernet0/1 from LOADING to FULL, Loading Done
Router#show ip ospf neighbor

Neighbor ID     Pri   State       Dead Time   Address        Interface
3.3.3.3         1    FULL/DR    00:00:38   10.10.30.2     GigabitEthernet0/1
2.2.2.2         1    FULL/DR    00:00:36   10.10.11.2     GigabitEthernet0/0
2.2.2.2         1    FULL/DR    00:00:59   10.10.10.2     GigabitEthernet0/3
Router#
```

Gambar 28: OSPF Neighbor dari Cisco Router

```
Neighbors

Area 0.0.0.0 interface 10.10.10.2 (Eth1/0/0)'s neighbors
Router ID: 1.1.1.1      Address: 10.10.10.1
State: Full            Mode:Nbr is Slave      Priority: 1
DR: 10.10.10.1        BDR: 10.10.10.2      MTU: 1500
Dead timer due in 38 sec
Retrans timer interval: 5
Neighbor is up for 00:00:34
Neighbor Up Time : 2026-06-05 03:20:50
Authentication Sequence: [ 0 ]

Area 0.0.0.0 interface 10.10.20.1 (Eth1/0/3)'s neighbors
Router ID: 3.3.3.3      Address: 10.10.20.2
State: Full            Mode:Nbr is Master     Priority: 1
DR: 10.10.20.1        BDR: 10.10.20.2      MTU: 1500
Dead timer due in 31 sec
Retrans timer interval: 5
Neighbor is up for 00:00:11
Neighbor Up Time : 2026-06-05 03:20:13
Authentication Sequence: [ 0 ]

Area 0.0.0.0 interface 10.10.11.2 (Eth1/0/4)'s neighbors
Router ID: 1.1.1.1      Address: 10.10.11.1
State: Full            Mode:Nbr is Slave      Priority: 1
DR: 10.10.11.1        BDR: 10.10.11.2      MTU: 1500
Dead timer due in 36 sec
Retrans timer interval: 5
Neighbor is up for 00:00:27
Neighbor Up Time : 2026-06-05 03:20:58
Authentication Sequence: [ 0 ]

[~HUAWEI]
```

Gambar 29: OSPF neighbor dari Huawei Router

```
[admin@Mikrotik] /> /routing ospf interface add interface-ether1 cost 100
[admin@Mikrotik] /> /routing ospf interface add interface-ether2 cost 100
[admin@Mikrotik] /> /routing ospf interface /
[admin@Mikrotik] /> /routing ospf neighbor print
0 instance-default router-id 1.1.1.1 address 10.10.30.1 interface-ether2 priority 1 dr-address 10.10.30.1
  backup-dr-address 10.10.30.2 state="Full" state-changes=5 ls-retransmits=0 ls-requests=0 db-summary=0
  adjacency=125
1 instance-default router-id 2.2.2.2 address 10.10.20.1 interface-ether1 priority 1 dr-address 10.10.20.1
  backup-dr-address 10.10.20.2 state="Full" state-changes=5 ls-retransmits=0 ls-requests=0 db-summary=0
  adjacency=125
[admin@Mikrotik] /> /ip route print where ospf
Flags: X - disabled, A - active, D - dynamic, C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
  0 - blackhole, U - unreachable, P - prohibit
# DST-ADDRESS      PREF-SRC      GATEWAY      DISTANCE
0 Ado 10.10.10.0/30      10.10.20.1      110
1 Ado 10.10.11.0/30      10.10.20.1      110
2 Ado 192.168.10.0/24     10.10.30.1      110
3 Ado 192.168.20.0/24     10.10.30.1      110
4 Ado 192.168.30.0/24     10.10.30.1      110
5 Ado 192.168.40.0/24     10.10.30.1      110
6 Ado 192.168.50.0/24     10.10.30.1      110
[admin@Mikrotik] /> /routing ospf neighbor print
0 instance-default router-id 1.1.1.1 address 10.10.30.1 interface-ether2 priority 1 dr-address 10.10.30.1
  backup-dr-address 10.10.30.2 state="Full" state-changes=5 ls-retransmits=0 ls-requests=0 db-summary=0
  adjacency=2m6s
1 instance-default router-id 2.2.2.2 address 10.10.20.1 interface-ether1 priority 1 dr-address 10.10.20.1
  backup-dr-address 10.10.20.2 state="Full" state-changes=5 ls-retransmits=0 ls-requests=0 db-summary=0
  adjacency=2m6s
[admin@Mikrotik] />
```

Gambar 30: OSPF neighbor dari Mikrotik

8. Verifikasi Routing Table

```
10.10.10.0/24 area 0
10.10.11.0/24 area 0
10.10.20.0/24 area 0
192.168.10.0/24 area 0
192.168.20.0/24 area 0
192.168.30.0/24 area 0
192.168.40.0/24 area 0
192.168.50.0/24 area 0
Routing Information Sources:
Gateway Distance Last Update
Distance: (default is 110)

Router#
*Jun 5 03:20:38.246: NSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on GigabitEthernet0/3 from LOADING to FULL, Loading Done
*Jun 5 03:20:45.549: NSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on GigabitEthernet0/9 from LOADING to FULL, Loading Done
Router#
*Jun 5 03:27:38.478: NSPF-5-ADJCHG: Process 1, Nbr 3.3.3.3 on GigabitEthernet0/1 from LOADING to FULL, Loading Done
*Jun 5 03:28:01.463: NSPF-5-ADJCHG: Process 1, Nbr 3.3.3.3 on GigabitEthernet0/1 from LOADING to FULL, Loading Done
Router#show ip ospf neighbor

Neighbor ID Pri State Dead Time Address Interface
3.3.3.3 1 FULL/BDR 00:00:38 10.10.30.2 GigabitEthernet0/1
2.2.2.2 1 FULL/BDR 00:00:36 10.10.11.2 GigabitEthernet0/9
2.2.2.2 1 FULL/BDR 00:00:29 10.10.10.2 GigabitEthernet0/3

Router#show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
Known via "ospf 1", distance 110, metric 11, type intra area
Last update from 10.10.11.2 on GigabitEthernet0/9, 00:09:34 ago
Routing Descriptor Blocks:
10.10.11.2, from 2.2.2.2, 00:09:34 ago, via GigabitEthernet0/9
Route metric is 11, traffic share count is 1
* 10.10.10.2, from 2.2.2.2, 00:09:48 ago, via GigabitEthernet0/3
Route metric is 11, traffic share count is 1
Router#
```

Gambar 31: Cisco Router

```
[~HUAWEI]display ip routing-table protocol ospf
Route Flags: R - relay, D - download to fib, T - to vpn-instance, B - black hole route
-----
..public.. Routing Table : OSPF
Destinations : 9 Routes : 14

OSPF routing table status : <Active>
Destinations : 5 Routes : 10

Destination/Mask Proto Pre Cost Flags NextHop Interface
10.10.30.0/30 OSPF 10 110 0 10.10.11.1 Ethernet1/0/4
10.10.30.0/30 OSPF 10 110 0 10.10.10.1 Ethernet1/0/9
192.168.10.0/24 OSPF 10 11 0 10.10.11.1 Ethernet1/0/4
192.168.10.0/24 OSPF 10 11 0 10.10.10.1 Ethernet1/0/9
192.168.20.0/24 OSPF 10 11 0 10.10.11.1 Ethernet1/0/4
192.168.20.0/24 OSPF 10 11 0 10.10.10.1 Ethernet1/0/9
192.168.30.0/24 OSPF 10 11 0 10.10.11.1 Ethernet1/0/4
192.168.30.0/24 OSPF 10 11 0 10.10.10.1 Ethernet1/0/9
192.168.40.0/24 OSPF 10 11 0 10.10.11.1 Ethernet1/0/4
192.168.40.0/24 OSPF 10 11 0 10.10.10.1 Ethernet1/0/9

OSPF routing table status : <Inactive>
Destinations : 4 Routes : 4

Destination/Mask Proto Pre Cost Flags NextHop Interface
10.10.10.0/30 OSPF 10 10 10.10.10.2 Ethernet1/0/9
10.10.11.0/30 OSPF 10 10 10.10.11.2 Ethernet1/0/4
10.10.20.0/30 OSPF 10 100 10.10.20.1 Ethernet1/0/3
192.168.50.0/24 OSPF 10 1 192.168.50.1 Ethernet1/0/2

[~HUAWEI]
```

Gambar 32: Huawei Router

```
B - blackhole, U - unreachable, P - prohibit
# DST-ADDRESS PREF-SRC GATEWAY DISTANCE
0 Ado 10.10.10.0/30 10.10.20.1 110
1 Ado 10.10.11.0/30 10.10.30.1 110
2 Ado 192.168.10.0/24 10.10.30.1 110
3 Ado 192.168.20.0/24 10.10.30.1 110
4 Ado 192.168.30.0/24 10.10.30.1 110
5 Ado 192.168.40.0/24 10.10.30.1 110
6 Ado 192.168.50.0/24 10.10.20.1 110

[admin@mikrotik] /> /routing ospf neighbor print
0 instance=default router-id=1.1.1.1 address=10.10.30.1 interface=ether2 priority=1 dr-address=10.10.30.1
backup-dr-address=10.10.30.2 state="Full" state-changes=5 ls-retransmits=0 ls-requests=0 db-summaries=0
adjacency=2ms

1 instance=default router-id=2.2.2.2 address=10.10.20.1 interface=ether1 priority=1 dr-address=10.10.20.1
backup-dr-address=10.10.20.2 state="Full" state-changes=5 ls-retransmits=0 ls-requests=0 db-summaries=0
adjacency=2ms

[admin@mikrotik] /> /ip route print where ospf
Flags: X - disabled, A - active, D - dynamic, C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
B - blackhole, U - unreachable, P - prohibit
# DST-ADDRESS PREF-SRC GATEWAY DISTANCE
0 Ado 10.10.10.0/30 10.10.20.1 110
1 Ado 10.10.11.0/30 10.10.30.1 110
2 Ado 192.168.10.0/24 10.10.30.1 110
3 Ado 192.168.20.0/24 10.10.30.1 110
4 Ado 192.168.30.0/24 10.10.30.1 110
5 Ado 192.168.40.0/24 10.10.30.1 110
6 Ado 192.168.50.0/24 10.10.20.1 110

[admin@mikrotik] />
```

Gambar 33: Hasil dari Mikrotik

9. Verifikasi Koneksi end-to-end

```
VPIC> ping 192.168.20.11
84 bytes from 192.168.20.11 icmp_seq=1 ttl=63 time=3.717 ms
84 bytes from 192.168.20.11 icmp_seq=2 ttl=63 time=3.631 ms
84 bytes from 192.168.20.11 icmp_seq=3 ttl=63 time=3.539 ms
84 bytes from 192.168.20.11 icmp_seq=4 ttl=63 time=3.960 ms
84 bytes from 192.168.20.11 icmp_seq=5 ttl=63 time=2.915 ms

VPIC> ping 192.168.30.10
84 bytes from 192.168.30.10 icmp_seq=1 ttl=63 time=1.927 ms
84 bytes from 192.168.30.10 icmp_seq=2 ttl=63 time=3.130 ms
84 bytes from 192.168.30.10 icmp_seq=3 ttl=63 time=2.809 ms
84 bytes from 192.168.30.10 icmp_seq=4 ttl=63 time=3.663 ms
84 bytes from 192.168.30.10 icmp_seq=5 ttl=63 time=3.007 ms

VPIC> ping 192.168.40.10
84 bytes from 192.168.40.10 icmp_seq=1 ttl=63 time=2.632 ms
84 bytes from 192.168.40.10 icmp_seq=2 ttl=63 time=2.809 ms
84 bytes from 192.168.40.10 icmp_seq=3 ttl=63 time=2.148 ms
84 bytes from 192.168.40.10 icmp_seq=4 ttl=63 time=3.836 ms
84 bytes from 192.168.40.10 icmp_seq=5 ttl=63 time=3.395 ms

VPIC> ping 192.168.50.10
84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=7.538 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=3.052 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=3.165 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=3.622 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=3.386 ms

VPIC>
```

Gambar 34: Ping dari PC VLAN 10

```
VRCS> ip dhcp
DOORA IP 192.168.20.11/24 GW 192.168.20.1

VRCS> show ip

NAME      : VPCS[1]
IP/MASK    : 192.168.20.11/24
GATEWAY    : 192.168.20.1
DNS        : 8.8.8.8
DHCP SERVER : 192.168.20.1
DHCP LEASE  : 86899, 86899/43200/75600
MAC        : 08:50:79:66:68:40
LPORT      : 20000
RHOST:PORT : 127.0.0.1:30000
MTU        : 1500

VRCS> ping 192.168.20.1

84 bytes from 192.168.20.1 icmp_seq=1 ttl=255 time=1.519 ms
84 bytes from 192.168.20.1 icmp_seq=2 ttl=255 time=1.774 ms
84 bytes from 192.168.20.1 icmp_seq=3 ttl=255 time=1.417 ms
84 bytes from 192.168.20.1 icmp_seq=4 ttl=255 time=1.513 ms
84 bytes from 192.168.20.1 icmp_seq=5 ttl=255 time=1.732 ms

VRCS> ping 192.168.50.10

84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=3.207 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=3.655 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=3.313 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=3.561 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=3.373 ms

VRCS>
```

Gambar 35: Ping dari PC VLAN 20

```
VRCS> ip 192.168.30.10/24 192.168.30.1
Checking for duplicate address
PCI : 192.168.30.10 255.255.255.0 gateway 192.168.30.1

VRCS> show ip

NAME      : VPCS[1]
IP/MASK    : 192.168.30.10/24
GATEWAY    : 192.168.30.1
DNS        :
MAC        : 08:50:79:66:68:3f
LPORT      : 20000
RHOST:PORT : 127.0.0.1:30000
MTU        : 1500

VRCS> ping 192.168.30.1

84 bytes from 192.168.30.1 icmp_seq=1 ttl=255 time=1.825 ms
84 bytes from 192.168.30.1 icmp_seq=2 ttl=255 time=1.485 ms
84 bytes from 192.168.30.1 icmp_seq=3 ttl=255 time=1.899 ms
84 bytes from 192.168.30.1 icmp_seq=4 ttl=255 time=1.746 ms
84 bytes from 192.168.30.1 icmp_seq=5 ttl=255 time=1.979 ms

VRCS> ping 192.168.50.10

84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=4.250 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=2.974 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=2.904 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=3.813 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=3.905 ms

VRCS>
```

Gambar 36: Ping dari PC VLAN 30

```
VRCS> ip 192.168.40.10/24 192.168.40.1
Checking for duplicate address...
*HPC1 : 192.168.40.10 255.255.255.0 gateway 192.168.40.1

VRCS> show ip

NAME      : VPCS[1]
IP/MASK    : 192.168.40.10/24
GATEWAY    : 192.168.40.1
DNS        :
MAC        : 08:50:79:66:68:3e
LPORT      : 20000
RHOST:PORT : 127.0.0.1:30000
MTU        : 1500

VRCS> ping 192.168.40.1

84 bytes from 192.168.40.1 icmp_seq=1 ttl=255 time=1.637 ms
84 bytes from 192.168.40.1 icmp_seq=2 ttl=255 time=1.731 ms
84 bytes from 192.168.40.1 icmp_seq=3 ttl=255 time=2.004 ms
84 bytes from 192.168.40.1 icmp_seq=4 ttl=255 time=1.394 ms
84 bytes from 192.168.40.1 icmp_seq=5 ttl=255 time=1.175 ms

VRCS> ping 192.168.50.10

84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=2.846 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=3.865 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=4.549 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=7.746 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=4.141 ms

VRCS>
```

Gambar 37: Ping dari PC VLAN 40

```
pc huawei> ping 192.168.10.11

84 bytes from 192.168.10.11 icmp_seq=1 ttl=62 time=3.906 ms
84 bytes from 192.168.10.11 icmp_seq=2 ttl=62 time=3.846 ms
84 bytes from 192.168.10.11 icmp_seq=3 ttl=62 time=3.975 ms
84 bytes from 192.168.10.11 icmp_seq=4 ttl=62 time=3.081 ms
84 bytes from 192.168.10.11 icmp_seq=5 ttl=62 time=4.138 ms

VRCS> ping 192.168.20.11

84 bytes from 192.168.20.11 icmp_seq=1 ttl=62 time=4.620 ms
84 bytes from 192.168.20.11 icmp_seq=2 ttl=62 time=4.977 ms
84 bytes from 192.168.20.11 icmp_seq=3 ttl=62 time=4.534 ms
84 bytes from 192.168.20.11 icmp_seq=4 ttl=62 time=3.633 ms
84 bytes from 192.168.20.11 icmp_seq=5 ttl=62 time=3.803 ms

VRCS> ping 192.168.30.10

84 bytes from 192.168.30.10 icmp_seq=1 ttl=62 time=3.231 ms
84 bytes from 192.168.30.10 icmp_seq=2 ttl=62 time=4.432 ms
84 bytes from 192.168.30.10 icmp_seq=3 ttl=62 time=4.670 ms
84 bytes from 192.168.30.10 icmp_seq=4 ttl=62 time=3.897 ms
84 bytes from 192.168.30.10 icmp_seq=5 ttl=62 time=4.630 ms

VRCS> ping 192.168.40.10

84 bytes from 192.168.40.10 icmp_seq=1 ttl=62 time=4.200 ms
84 bytes from 192.168.40.10 icmp_seq=2 ttl=62 time=2.802 ms
84 bytes from 192.168.40.10 icmp_seq=3 ttl=62 time=3.721 ms
84 bytes from 192.168.40.10 icmp_seq=4 ttl=62 time=6.683 ms
84 bytes from 192.168.40.10 icmp_seq=5 ttl=62 time=4.352 ms

VRCS>
```

Gambar 38: Ping dari PC LAN huawei ke IP VLAN Cisco

1.6 Praktikum 5 : Pengujian Failover OSPF

1. Kondisi Normal : Dua link Cisco-Huawei aktif

```
Route metric is 11, traffic share count is 1
+ 10.10.10.2, from 2.2.2.2, 00:20:11 ago, via GigabitEthernet0/3
Route metric is 11, traffic share count is 1
Router#tracert 192.168.50.10
Type escape sequence to abort.
Tracing the route to 192.168.50.10
VRF info: (vrf in name/id, vrf out name/id)
 0 10.10.10.2 5 msec
 1 10.10.11.2 1 msec
 2 10.10.10.2 2 msec
 3 * * *
 4 * * *
 5 * * *
 6 * * *
 7 * * *
 8 * * *
 9 * * *
10 * * *
11 * * *
12 * * *
13 * * *
14 * * *
15 * * *
16 * * *
17 * * *
18 * * *
19 * * *
20 * * *
21 * * *
22 * * *
23 * * *
24 * * *
25 * * *
26 * * *
27 * * *
28 * * *
29 * * *
30 * * *
31 * * *
32 * * *
33 * * *
34 * * *
35 * * *
36 * * *
37 * * *
38 * * *
39 * * *
40 * * *
41 * * *
42 * * *
43 * * *
44 * * *
45 * * *
Router#tracert 192.168.50.10
Type escape sequence to abort.
Tracing the route to 192.168.50.10
VRF info: (vrf in name/id, vrf out name/id)
 0 10.10.10.2 3 msec
 1 10.10.11.2 1 msec
 2 10.10.10.2 1 msec
 3 192.168.50.10 7 msec 2 msec 2 msec
Router#
```

Gambar 39: Route dari Cisco menuju LAN Huawei

2. Kondisi 1 Link Cisco-Huawei mati

```
NAME      : VPCS[1]
IP/PAUSE  : 192.168.30.18/24
GATEWAY   : 192.168.30.1
DNS       :
MAC       : 08:50:79:66:68:3f
LPORT     : 20000
RHOST.PORT : 127.0.0.1:30000
MTU       : 1500

VPCS> ping 192.168.30.1
84 bytes from 192.168.30.1 icmp_seq=1 ttl=255 time=1.825 ms
84 bytes from 192.168.30.1 icmp_seq=2 ttl=255 time=1.485 ms
84 bytes from 192.168.30.1 icmp_seq=3 ttl=255 time=1.899 ms
84 bytes from 192.168.30.1 icmp_seq=4 ttl=255 time=1.746 ms
84 bytes from 192.168.30.1 icmp_seq=5 ttl=255 time=1.979 ms

VPCS> ping 192.168.50.10
84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=4.258 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=2.974 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=2.944 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=3.613 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=3.905 ms

VPCS> trace 192.168.50.10
Trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1 192.168.30.1 4.383 ms 2.333 ms 2.456 ms
 2 10.10.10.2 4.367 ms 2.418 ms 3.359 ms
 3 *192.168.50.10 3.744 ms (ICMP type=3, code=3, Destination port unreachable)

VPCS>
```

Gambar 40: Link Cisco-Huawei Mati

3. Kondisi 2 Link Cisco-Huawei Mati

```
LPORT     : 20000
RHOST.PORT : 127.0.0.1:30000
MTU       : 1500

VPCS> ping 192.168.30.1
84 bytes from 192.168.30.1 icmp_seq=1 ttl=255 time=1.825 ms
84 bytes from 192.168.30.1 icmp_seq=2 ttl=255 time=1.485 ms
84 bytes from 192.168.30.1 icmp_seq=3 ttl=255 time=1.899 ms
84 bytes from 192.168.30.1 icmp_seq=4 ttl=255 time=1.746 ms
84 bytes from 192.168.30.1 icmp_seq=5 ttl=255 time=1.979 ms

VPCS> ping 192.168.50.10
84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=4.258 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=2.974 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=2.944 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=3.613 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=3.905 ms

VPCS> trace 192.168.50.10
Trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1 192.168.30.1 4.383 ms 2.333 ms 2.456 ms
 2 10.10.10.2 4.367 ms 2.418 ms 3.359 ms
 3 *192.168.50.10 3.744 ms (ICMP type=3, code=3, Destination port unreachable)

VPCS> trace 192.168.50.10
Trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1 192.168.30.1 1.928 ms 1.638 ms 3.372 ms
 2 10.10.11.2 2.723 ms 3.551 ms 2.538 ms
 3 *192.168.50.10 5.579 ms (ICMP type=3, code=3, Destination port unreachable)

VPCS>
```

Gambar 41: Kondisi 2 Mati

4. Mengaktifkan kembali dua link cisco-huawei

```
84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=4.258 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=2.974 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=2.944 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=3.613 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=3.905 ms

VPCS> trace 192.168.50.10
Trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1 192.168.30.1 4.383 ms 2.333 ms 2.456 ms
 2 10.10.11.2 2.723 ms 3.551 ms 2.538 ms
 3 *192.168.50.10 3.744 ms (ICMP type=3, code=3, Destination port unreachable)

VPCS> trace 192.168.50.10
Trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1 192.168.30.1 1.928 ms 1.638 ms 3.372 ms
 2 10.10.11.2 2.723 ms 3.551 ms 2.538 ms
 3 *192.168.50.10 5.579 ms (ICMP type=3, code=3, Destination port unreachable)

VPCS> trace 192.168.50.10
Trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1 192.168.30.1 3.862 ms 2.728 ms 4.076 ms
 2 10.10.38.2 3.423 ms 4.426 ms 5.813 ms
 3 10.10.20.1 6.838 ms 6.405 ms 2.458 ms
 4 *192.168.50.10 3.976 ms (ICMP type=3, code=3, Destination port unreachable)

VPCS> trace 192.168.50.10
Trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1 192.168.30.1 2.402 ms 2.111 ms 2.077 ms
 2 10.10.10.2 2.911 ms 1.594 ms 2.265 ms
 3 *192.168.50.10 3.728 ms (ICMP type=3, code=3, Destination port unreachable)

VPCS>
```

Gambar 42: Jalur melewati Mikrotik

2 Analisis Hasil Percobaan

Pada praktikum ini, dilakukan konfigurasi jaringan multivendor yang melibatkan Cisco Switch, Cisco Router, Huawei Router, dan MikroTik RouterOS, dengan implementasi VLAN, trunking, DHCP, dan routing dinamis OSPF. Pada Praktikum 1, Cisco Switch dikonfigurasi dengan empat VLAN, yaitu VLAN 10 (FINANCE), VLAN 20 (HR), VLAN 30 (IT), dan VLAN 40 (MARKETING), yang seluruhnya berhasil dibuat dan berstatus aktif. Selanjutnya, port akses dikonfigurasi sesuai VLAN masing-masing, sementara koneksi ke Cisco Router dikonfigurasi sebagai trunk dengan encapsulation 802.1Q dan mengizinkan keempat VLAN melewati link tersebut, yang dapat diverifikasi melalui perintah `show interfaces trunk`.

Pada Praktikum 2, Cisco Router dikonfigurasi dengan sub-interface trunk untuk setiap VLAN (GigabitEthernet0/2.10, 0/2.20, 0/2.30, dan 0/2.40), masing-masing diberi alamat gateway 192.168.X0.1/24, serta layanan DHCP server untuk setiap VLAN. Hasil verifikasi menunjukkan seluruh interface berstatus up dengan protokol up, sedangkan pool DHCP untuk VLAN 10 dan VLAN 20 menunjukkan utilization rate 100% dengan satu alamat ter-lease, dan binding IP DHCP mencatat alamat 192.168.10.11 dan 192.168.20.11 berhasil disewakan secara otomatis ke client. Pengujian dari sisi VPCS menunjukkan PC pada VLAN 10, 20, 30, dan 40 berhasil memperoleh IP address secara DHCP maupun statis sesuai subnet masing-masing, dan seluruhnya berhasil melakukan ping ke gateway dengan round-trip time berkisar 1–5 ms tanpa packet loss. Pengujian ping antar-VLAN dari PC VLAN 10 ke gateway VLAN 20, 30, dan 40 juga menunjukkan keberhasilan penuh (0% loss), yang membuktikan bahwa fungsi inter-VLAN routing pada Cisco Router telah berjalan dengan benar.

Pada Praktikum 3, dilakukan konfigurasi LAN Huawei dengan subnet 192.168.50.0/24 serta interlink antara Cisco–Huawei (dua link) dan Huawei–MikroTik menggunakan subnet point-to-point /30 (10.10.10.0/30, 10.10.11.0/30, dan 10.10.20.0/30, 10.10.30.0/30). Verifikasi interface pada ketiga perangkat menunjukkan seluruh link berstatus up baik secara fisik maupun protokol. Pengujian ping antar router (Cisco ke Huawei, Huawei ke Cisco dan MikroTik, serta MikroTik ke Huawei) menghasilkan success rate 80–100% dengan RTT 1–2 ms, mengonfirmasi konektivitas Layer 3 antar perangkat multivendor telah terbentuk dengan baik.

Pada Praktikum 4, OSPF dikonfigurasi pada keempat perangkat (Cisco, Huawei, dan MikroTik) dengan area 0.0.0.0 (backbone). Hasil verifikasi pada Cisco Router menunjukkan tiga neighbor OSPF dalam status FULL (Router ID 3.3.3.3, 2.2.2.2, dan 2.2.2.3), sedangkan pada Huawei Router teridentifikasi dua neighbor dalam status Full dengan role Master maupun Slave (DR/BDR election berjalan normal). Pada MikroTik, dua adjacency OSPF (Router ID default dan 2.2.2.2) juga berstatus Full. Routing table pada masing-masing router menunjukkan bahwa rute-rute menuju subnet VLAN (192.168.10.0/24 hingga 192.168.40.0/24) dan LAN Huawei (192.168.50.0/24) berhasil dipelajari melalui OSPF dengan cost yang konsisten (cost 10–110), serta route OSPF inter-area maupun intra-area (O dan IA) muncul pada Huawei sesuai dengan topologi multi-hop yang ada. Pengujian konektivitas end-to-end dari PC VLAN 10, 20, 30, dan 40 menuju subnet LAN Huawei (192.168.50.0/24), serta sebaliknya dari PC LAN Huawei menuju subnet VLAN Cisco, seluruhnya berhasil dengan RTT berkisar 1–9 ms tanpa packet loss, membuktikan bahwa OSPF telah berhasil mendistribusikan informasi routing secara menyeluruh pada jaringan multivendor ini.

Pada Praktikum 5, dilakukan pengujian failover OSPF dengan memutus link Cisco–Huawei secara bertahap. Pada kondisi normal (dua link aktif), trace route dari Cisco menuju LAN Huawei (192.168.50.0/24) melewati jalur langsung via GigabitEthernet0/3 dengan traffic share count bernilai

1, menunjukkan adanya load balancing antar dua link dengan cost yang sama (equal-cost multipath). Ketika satu link Cisco–Huawei diputus, trace route menunjukkan paket tetap dapat mencapai tujuan melalui jalur alternatif (link kedua), meskipun terjadi peningkatan jumlah hop dan delay menjadi sekitar 3,3–5,6 ms. Pada kondisi kedua link Cisco–Huawei mati secara bersamaan, trace route menunjukkan paket dialihkan melalui jalur melewati MikroTik (10.10.30.0/30) dengan tambahan hop dan delay yang lebih tinggi (hingga 6,5 ms), namun tetap dapat mencapai tujuan akhir (meskipun pada beberapa percobaan muncul *ICMP Destination port unreachable* pada hop terakhir karena traceroute menggunakan UDP probe, bukan indikasi kegagalan routing). Setelah kedua link Cisco–Huawei diaktifkan kembali, OSPF kembali melakukan reconvergence dan jalur kembali stabil. Hasil ini menunjukkan bahwa mekanisme failover OSPF berjalan dengan baik, di mana OSPF secara otomatis melakukan rekalkulasi SPF (Shortest Path First) dan mengalihkan trafik ke jalur alternatif ketika terjadi kegagalan link, tanpa menyebabkan putusnya konektivitas secara permanen.

3 Kesimpulan

Berdasarkan hasil percobaan yang telah dilakukan, dapat disimpulkan bahwa konfigurasi VLAN, trunking, DHCP, dan OSPF pada jaringan multivendor yang terdiri dari Cisco, Huawei, dan MikroTik dapat berjalan dengan baik dan saling terintegrasi meskipun masing-masing perangkat memiliki sintaks konfigurasi yang berbeda. Segmentasi jaringan menggunakan VLAN berhasil memisahkan empat departemen (FINANCE, HR, IT, dan MARKETING) ke dalam broadcast domain yang terpisah, sementara konfigurasi trunk 802.1Q memungkinkan seluruh VLAN tersebut dilewatkan melalui satu link fisik antara switch dan router. Layanan DHCP server pada Cisco Router juga berhasil mengalokasikan IP address secara otomatis kepada client di setiap VLAN, yang dibuktikan dengan keberhasilan binding dan ping ke gateway. Lebih lanjut, OSPF sebagai protokol routing dinamis terbukti mampu beroperasi secara multivendor, membentuk neighbor adjacency dalam status Full antar Cisco, Huawei, dan MikroTik, serta berhasil mendistribusikan informasi routing untuk seluruh subnet pada jaringan, sehingga konektivitas end-to-end antara seluruh VLAN dan LAN Huawei dapat tercapai. Selain itu, pengujian failover menunjukkan bahwa OSPF memiliki kemampuan self-healing yang baik, di mana ketika terjadi kegagalan pada satu atau bahkan dua link sekaligus, trafik secara otomatis dialihkan melalui jalur alternatif (termasuk melalui MikroTik) tanpa intervensi manual, sehingga konektivitas jaringan tetap terjaga meskipun dengan delay yang sedikit lebih tinggi. Secara keseluruhan, praktikum ini berhasil mendemonstrasikan bahwa interoperabilitas antar vendor jaringan dapat diwujudkan melalui penggunaan protokol standar seperti 802.1Q dan OSPF, serta menunjukkan pentingnya redundansi link dalam menjaga ketersediaan (availability) jaringan.

4 Lampiran

4.1 Dokumentasi saat praktikum



Gambar 43: Foto waktu pengerjaan